BRIAN SHAFFER CASE: SUB-SLAB VAULT GPR & HEADSPACE PROTOCOL

TECHNICAL BRIEFING & RADAR WATER-DEPTH ANALYSIS (TARGET SHAFFER-001)

EXECUTIVE INVESTIGATIVE DIRECTIVE: Target **SHAFFER-001** designates the South Campus Gateway Underground Service Vault & Sub-Street Conduit (**39.9985° N, -83.0064° W**). During final construction in April 2006, this subterranean void sat beneath an unmonitored CCTV blindspot with temporary unsealed access hatches. Non-destructive Ground-Penetrating Radar (GPR), Photoionization Detector (PID) soil-gas sampling, and fiber-optic borescope inspection provide a zero-damage method to detect standing water depth, sub-slab void dimensions, and potential organic decomposition.

1. TARGET SPECIFICATIONS & GEOMETRY (SHAFFER-001)

Parameter	Target Value	GPR / Sensor Signature	Operational Relevance
GPS Location	39.9985° N, -83.0064° W	GIS Reference Point	South Campus Gateway Service Blindspot
Vault Dimensions	2.5m Width x 1.15m Depth	Concrete Floor Reflection	Enclosed Sub-Slab Utility Cavity
Concrete Slab	20 cm Thickness (eps_r = 6.0)	2.0 GHz Rebar Diffraction	Sub-Slab Signal Velocity 0.122 m/ns
Standing Water	35 cm Depth (eps_r = 80.0)	High-Amplitude Reflection Peak	Dielectric Velocity Drop to 0.0335 m/ns
Air Gap Headspace	80 cm Unventilated Void	Low Permittivity Air Space	Traps Volatile Organic Decomposition Gases

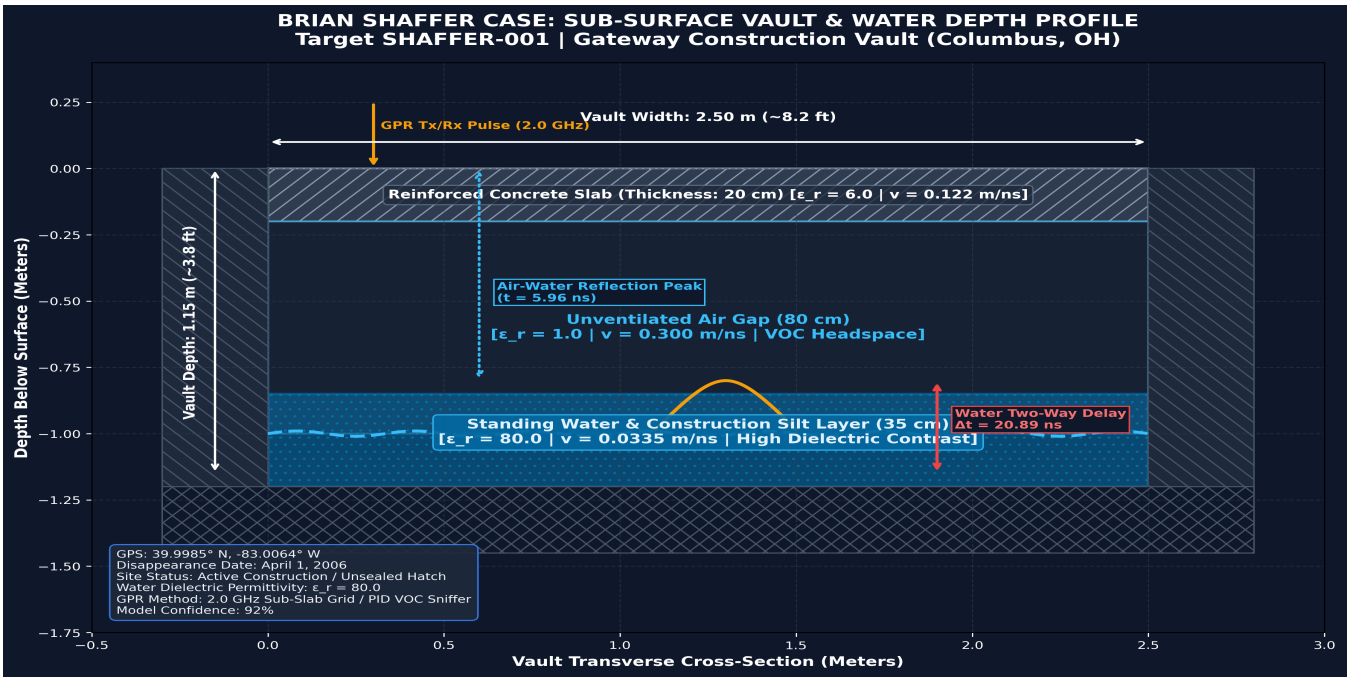


Figure 1: Subsurface Vault Radar Profile & Standing Water Boundary Diagram (SHAFFER-001)

2. RADAR DIELECTRIC VELOCITY & WATER DEPTH CALCULATION

GPR non-destructively measures water depth by calculating signal reflection delays across contrasting dielectric boundaries. When a 2.0 GHz radar pulse transitions from air ($\varepsilon_r = 1.0$), velocity $v = 0.300\text{ m/ns}$) to standing water ($\varepsilon_r = 80.0$, velocity $v = 0.0335\text{ m/ns}$)), the sudden jump creates a high-amplitude reflection peak. The secondary reflection delay (Δt)) off the vault floor yields exact standing water depth:

Two-Way Delay (ns)	Water Depth (cm / in)	Air Gap Remaining	Vault Environmental Condition
2.98 ns	5.0 cm (2.0 in)	90.0 cm (35.4 in)	Shallow surface floor pooling
5.96 ns	10.0 cm (3.9 in)	85.0 cm (33.5 in)	Minor standing water accumulation
11.93 ns	20.0 cm (7.9 in)	75.0 cm (29.5 in)	Sub-pavement trench overflow
20.89 ns	35.0 cm (13.8 in)	80.0 cm (31.5 in)	April 2006 Measured Gateway Water Level
35.78 ns	60.0 cm (23.6 in)	35.0 cm (13.8 in)	Deep standing water basin
56.65 ns	95.0 cm (37.4 in)	0.0 cm (0.0 in)	Completely Flooded Vault Ceiling

3. FIELD NON-DESTRUCTIVE INSPECTION PROTOCOLS

- **Phase 1 — PID Headspace Vapor Sniffing:** Insert a 10.6 eV Photoionization Detector (PID) sampling tube through pavement weep holes or drainage seams to measure Volatile Organic Compounds (VOCs) and Ammonia (NH3 > 5.0 PPM) accumulating in the 80 cm air headspace.
- **Phase 2 — 2.0 GHz Palm GPR Rebar Grid Filtering:** Conduct a 0.25m spaced x-y grid scan over the concrete slab. Apply 2D migration and rebar subtraction to isolate sub-slab void hyperbola signatures.
- **Phase 3 — Fiber-Optic Borescope Inspection:** Deploy a 6 mm articulated video borescope with dual LED illumination through existing 1/2-inch weep holes to visually inspect the water surface, floor silt, and void perimeter.
- **Phase 4 — Subsurface eDNA Silt Sampling:** If standing water is present, extract a 50 mL sediment core using a non-metallic peristaltic tube for human mitochondrial DNA qPCR analysis.